

DATA SCIENCE CAPSTONE PROJECT

Phase 2

Data Understanding & Requirement Analysis

A complete end-to-end guide for translating business problems into analytical requirements, success criteria, and structured data plans.

■ Objective

The primary goal of Phase 2 is to **bridge the gap between the business problem (identified in Phase 1) and the technical data science solution**. This phase ensures that every stakeholder — business owners, data scientists, and developers — shares a common, unambiguous understanding of what the project must accomplish, what data is needed, and how success will be measured.

Without a thorough requirements analysis, projects risk building the wrong models, missing critical data, or delivering outputs that don't actually solve the business pain.

■ Why This Phase Matters

Risk / Gap	How Phase 2 Prevents It
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Building the wrong model	Clear business requirements pin down what to predict/analyze
Poor feature selection	Data requirements specify exactly which columns are needed
Ambiguous project success	Success metrics define measurable thresholds upfront
Scope creep	Documented constraints keep the project bounded
Miscommunication with stakeholders	Requirement document acts as a shared contract

■ Activity 1 — Business Requirements

Business requirements answer the **"why"** and **"what"** of the project. They translate domain pain points into precise, data-actionable questions that the data science solution must address.

1a. Decisions That Need to Be Improved

Start by listing specific decisions that currently rely on intuition, manual effort, or legacy reports. Each decision becomes a candidate for data-driven improvement.

- Which customers are likely to churn in the next 30 days?
- What will product demand be for the next quarter?
- Which loan applicants pose a high default risk?
- What price point maximises revenue for a product segment?
- Which students are at risk of dropping out this semester?

1b. Insights or Predictions Required

For each decision, define whether the output is a **prediction** (a number or class the model outputs) or an **insight** (a pattern or trend surfaced through analysis).

- Prediction: Probability score (0–1) of customer churn
- Insight: Top 3 factors driving high cart-abandonment rate
- Prediction: Next-month sales volume per product category
- Insight: Correlation between study hours and exam scores
- Prediction: Binary flag — fraudulent transaction (Yes/No)

1c. Expected Business Impact

Quantify the anticipated value that the solution will deliver. This justifies the project and anchors success metrics in real-world outcomes.

- Reduce churn-related revenue loss by 15% in Q3
- Cut manual reporting hours from 40 hrs/week to under 5
- Improve demand-forecast accuracy from 65% to 85%
- Increase early intervention rate for at-risk students by 30%
- Decrease false-positive fraud flags by 25%, saving analyst time

■■ Activity 2 — Data Requirements

Data requirements specify **exactly what data the project needs** to deliver the insights or predictions defined above. This section prevents the costly mistake of discovering missing data after model development has begun.

2a. Required Datasets

List every dataset that the project will need, whether already available or to be sourced.

- Customer transaction history (12 months minimum)
- Product inventory and sales records
- User demographic and behavioural data
- Sensor / IoT readings (for manufacturing use-cases)
- External economic indicators (GDP, inflation, UPI index)
- Public datasets from Kaggle, UCI ML Repository, data.gov

2b. Features (Columns) Needed

Identify specific variables — both **input features** for the model and the **target variable** it must predict or explain.

- Input features: age, income, transaction frequency, average order value
- Categorical: gender, region, product category, payment method
- Time-based: transaction date, recency, seasonality flags
- Derived features: days since last purchase, rolling 7-day spend
- Target variable: churn_flag (1=churned, 0=retained)

2c. Data Sources

Document where each dataset will come from — this informs feasibility and timelines.

- Internal database / ERP system exports (CSV or SQL query)
- Kaggle public datasets (kaggle.com/datasets)
- UCI Machine Learning Repository
- Government open data portals (data.gov.in, data.gov)
- Synthetic data generated via Faker / SDV / Gretel.ai
- Web scraping (with appropriate permissions)

2d. Data Frequency & Volume

Specify how often data is refreshed and its approximate scale — critical for pipeline design and compute planning.

- Historical (batch): 2–3 years of records, ~500K rows
- Near-real-time: hourly transaction feeds via API
- Static reference: product catalogue updated monthly
- Estimated volume: 50 MB – 2 GB (student/capstone scope)
- Streaming (advanced): Kafka topic for real-time predictions

■ Activity 3 — Success Metrics

Success metrics answer: **"How will we know when the model is good enough?"** They must be defined before training begins — not after — to prevent cherry-picking favourable results. Metrics are split into two categories: **Technical** (model performance) and **Business** (real-world impact).

3a. Technical / Model Metrics

Metric	Use-Case Type	Threshold Example
Accuracy	Classification (balanced)	$\geq 85\%$
Precision	When false positives are costly	$\geq 80\%$
Recall	When false negatives are costly	$\geq 78\%$
F1-Score	Imbalanced classification	≥ 0.80
AUC-ROC	Binary classifiers	≥ 0.85
RMSE	Regression / forecasting	$< 5\%$ of range
MAE	Regression / demand forecast	< 10 units

MAPE	Time-series forecasting	< 10%
Silhouette Score	Clustering	> 0.50

3b. Business KPIs (Key Performance Indicators)

Business KPIs link model performance to tangible outcomes that stakeholders care about.

KPI Category	Example Target
Revenue Impact	Reduce churn-driven revenue loss by █5L/quarter
Operational Efficiency	Cut manual data-entry hours from 40 to < 5 per week
Forecast Accuracy	Improve demand forecast accuracy from 65% → 85%
Fraud Detection Rate	Catch 90% of fraudulent transactions with < 5% false positives
User Adoption	80% of analysts actively using the dashboard within 4 weeks
Cost Savings	Reduce inventory over-stocking costs by 20%
Decision Speed	Reduce time-to-decision from 3 days to 4 hours

█ Deliverables

Deliverable	Contents	Format
Requirements Document	Business requirements, data requirements, stakeholder mapping, scope boundaries	Word / PDF, 3–5 pages
Success Metrics Definition	Technical metrics with thresholds, business KPIs with targets, baseline benchmarks	Table in doc / spreadsheet
Data Source Inventory	List of datasets, source URLs, access method, size estimate, refresh frequency	Spreadsheet / table
Analytical Questions List	Prioritised list of questions the data science solution must answer	Numbered list in doc

■ Practical Example — Budget Buddy (Group 13)

Below is a filled-out Phase 2 analysis for the Budget Buddy capstone project — an AI-powered personal finance manager for students and young professionals in India.

Business Requirements

Decision to Improve	Identify overspending patterns and predict future budget shortfalls
Insight Required	Category-wise spending trends; anomaly detection on UPI transactions
Prediction Required	Monthly budget forecast using Prophet + XGBoost models
Business Impact	Help users save 10–20% more per month; reduce financial anxiety

Data Requirements

Primary Dataset	Synthetic UPI transaction data (Faker + SDV); Kaggle Indian expense datasets
Key Features	transaction_date, amount, category, merchant, payment_method, user_id
Target Variable	budget_exceeded_flag (binary), next_month_spending (regression)
Data Volume	~100K synthetic rows + 50K real-world Kaggle records
Refresh Frequency	Real-time via voice/text input; batch re-training weekly

Success Metrics

Classification Metric	F1-Score ≥ 0.82 for budget-exceeded prediction
Regression Metric	MAPE $< 8\%$ for monthly spend forecasting
RAG Assistant Quality	User satisfaction $\geq 4.0/5$ in pilot testing
Business KPI	30% reduction in manual expense tracking effort
Engagement KPI	70% of users complete weekly financial check-in

■■ Common Mistakes to Avoid

X Vague requirements

"Improve predictions" is not a requirement. Specify what to predict, for whom, with what threshold.

X Skipping data inventory

Discovering a key dataset is unavailable mid-project is a project killer. Verify access upfront.

X No baseline defined

Without a baseline, you cannot claim improvement. Document current accuracy or decision speed.

X Ignoring data quality

Assume data is dirty until proven otherwise. Note expected null rates, outlier ranges, and encoding issues.

X Business-IT mismatch

Technical metrics alone don't satisfy stakeholders. Always pair model metrics with business KPIs.

■ Phase 2 Completion Checklist

- Business requirements documented (decisions, insights, predictions, impact)
- Data requirements specified (datasets, features, target variable, sources, volume)
- Success metrics defined with numeric thresholds (technical + business KPIs)
- Data source availability confirmed (access verified, not just listed)
- Baseline performance documented (current state before the model)
- Stakeholder sign-off obtained on requirements document
- Analytical questions list finalised and prioritised
- Scope boundaries stated (what is OUT of scope for this phase)

■ Phase Flow Overview

